

Tech Transcends Tongues

Machine Translation remains a tough problem for Artificial Intelligence. Can we expect more accurate systems in the future?

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“Data-oriented translation,” “Example-based and context-based processing,” “Syntactic parsing,” “Japanese morphological analysis,” and such is the stuff of machine translation (MT) research. But that’s exactly *not* what we’re going to talk about here, or even translate for you into English! Here, we take a gentle look at the potential of MT, and what we expect from it in the near future. Will you be able to shoot off an e-mail to a Spanish friend with it being translated in transit?

Your first brush with translation has probably been in the context of reverse translation using online translators such as Google’s and AltaVista’s. We’re talking about the forwards that go round saying something like “Translate the following to Spanish on Google and then translate it back into English! Have fun!” Reverse translation may be fun, but it’s unfair. Unfair, because it’s stressing the system too much, as it were.

But what about simple, one-way translation? We have the all-time classic of hilarious MT in the apocryphal story of a 1950s system that translated “The spirit is willing, but the flesh is weak” into Russian as “The vodka is strong, but the meat is rotten.” Now that might be a joke, or what, we don’t know, but let’s take a look at the state of the art: at www.systransoft.com, when you translate the German “Du Strahlenreiche” (You radiant one) into English, out comes “You jet realms.” And “Gleichnis” (likeness, or reflection) becomes “equalsneeze”!

Rays To Riches

That’s a sad state of affairs, but a good starting point for an analysis. “Strahlen” means rays or beams—hence “jet.” And “reich” means “rich,” but “Reich” can also mean “rule,” or “domain”—hence “realm.” Now, “Du Strahlenreiche,” for us, is a combination of “rays” and “rich,” making for a literal translation of “You who are rich in rays,” which we understand as “You radiant one.” But how is the poor software to understand that we

meant “rich” and not “realm”? More importantly, we gather from the context, and from our world knowledge, that “rich in rays” can mean “radiant.” How on earth would the software know that there is even any such thing as radiance in the world, given only the words “realm,” “rich,” and “rays”?

An analysis of some ambiguous text shows just how much a program needs to know in order to be able to faithfully render a piece. Consider three sentences -

“I like the cat, I can drink milk.”

“I, like the cat, can drink milk.”

“I, cat-like, can drink milk.”

“I can drink milk like the cat.”

Even supplied with all three meanings of “like,” a machine would find it really hard to translate all four sentences properly! Then take the example of “The shirt is in the trunk.” The storage trunk? An elephant’s trunk? A tree-trunk? We know that it’s the first one; why? Because an elephant’s trunk is too small to shove a shirt into, and because a tree-trunk is a highly unlikely place to stash away a shirt in! And who is to take on the job of feeding all world-knowledge into a translating machine? Then there’s idioms: “Woe is me” will, in all likeness, get translated as “I am sadness” into other languages.

We’re Tough Customers

As things around us develop at the proverbial mind-numbing pace, we’ve become impatient. We want IMs that translate, so you can type in English and your friend somewhere reads your text in Hebrew. We want a button on Web pages where we choose the language to view the page in. We want automatic speech translators, so we can speak in Hindi with our speech coming out in English through a mouthpiece.

Turns out something like that has already been developed, but more on that later. For now, have you realised how fortunate you are to be able to read this copy of *Digit*? You have access to more Web pages than most other people! Figures vary from source to source, but apart from the fact that Chinese is competing fast and hard, English is still by and far the most prevalent



Illustration: Shrikrishna Pakar